DTC P2502:00 [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

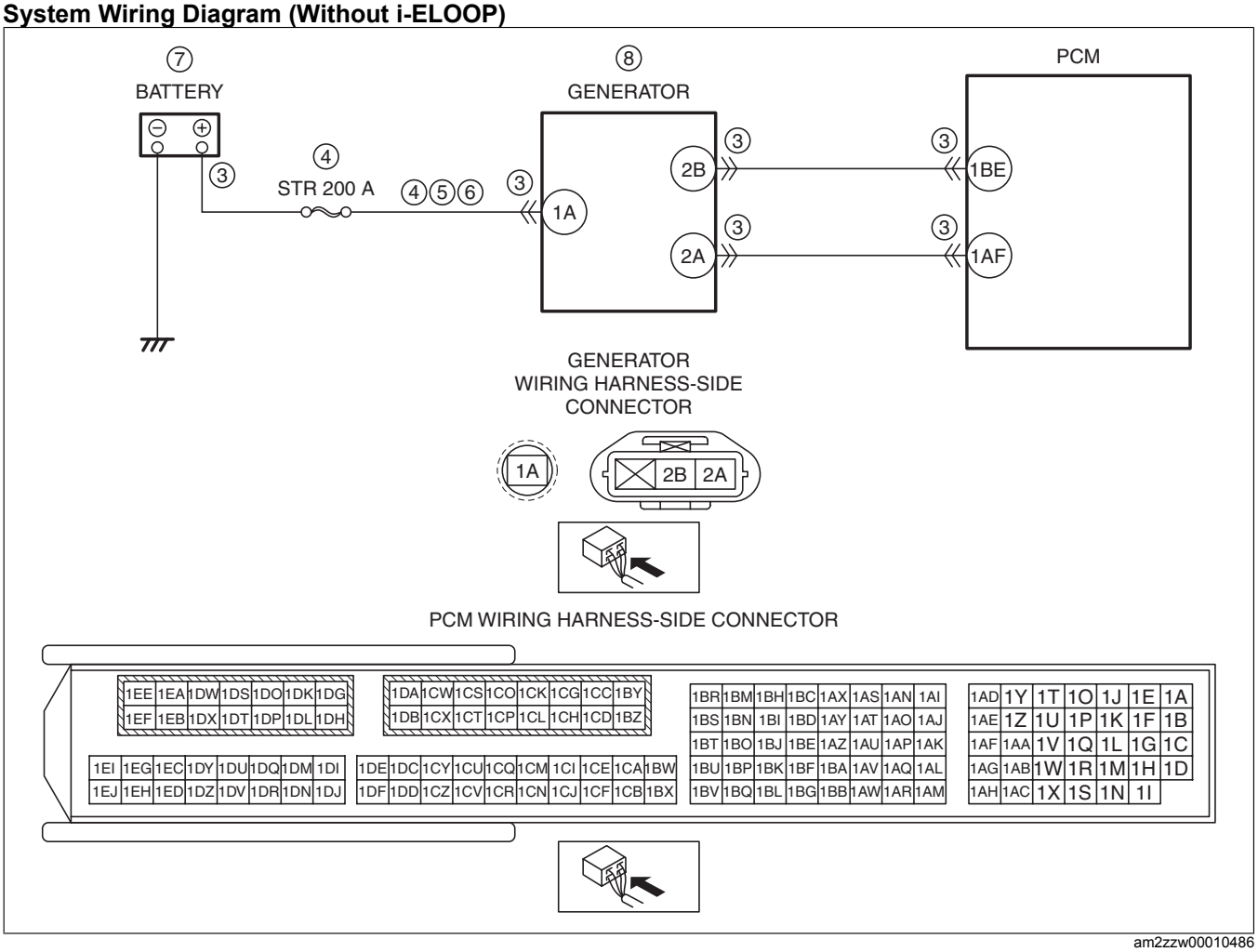
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Note
- To determine the malfunctioning part, proceed with the diagnostics from "Function Inspection Using M-MDS".

Details On DTCs (Without i-ELOOP)

DESCRIPTION	Generator system: Malfunction in voltage generated by generator	
DETECTION CONDITION	Determination conditions	• The voltage generated by the generator is **17 V or higher** and the battery voltage is **11 V or less** for a continuous specified time.
	Preconditions	• While engine is running
	Malfunction determination period	• **5 s** period
	Drive cycle	• 1
	Self test type	• CMDTC self test
	Sensor used	• PCM • Generator
FAIL-SAFE FUNCTION	• Inhibits engine-stop by operating the i-stop function. (With i-stop) • Generator control is inhibited.	
VEHICLE STATUS WHEN DTCs ARE OUTPUT	• Flashes i-stop warning light (amber). (With i-stop) • A warning message is displayed on the center display. (With center display) • Illuminates charging system warning light. • If the vehicle continues to be driven while the DTC is detected the battery will be depleted. — A malfunction occurs with an electrical device and the vehicle stops. • The following vehicle conditions differ depending on the type of malfunction: — Vehicle shock may occur due to generator load. — Idling feel due to generator-stop may occur.	
POSSIBLE CAUSE	• Poor connection of the following parts: — Battery — Generator — PCM • Connector or terminal malfunction of the following parts: — Battery — Generator — PCM • Short to ground or open circuit in generator charge/discharge circuit — Short to ground in wiring harness between battery positive terminal and generator terminal 1A — STR 200 A fuse malfunction — Open circuit in wiring harness between battery positive terminal and generator terminal 1A • Generator malfunction • Battery malfunction • PCM malfunction	

System Wiring Diagram (Without i-ELOOP)



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Function Explanation (DTC Detection Outline) (Without i-ELOOP)

- When the charge/discharge circuit for the power supplying the vehicle is normal, the vehicle will operate normally. In this diagnostic, a low vehicle supply voltage is detected even though the generator is generating power, a malfunction in the charge/discharge circuit is detected, and verification of vehicle malfunctions/safety assurance is performed.

Repeatability Verification Procedure (Without i-ELOOP)

1. Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
2. Start the engine.
3. Leave for **180 s** while idling with no electrical load.
4. Leave for **30 s** while idling with high electrical load.

PID Item/Simulation Item Used In Diagnosis (Without i-ELOOP)

- Not applicable

Function Inspection Using M-MDS (Without i-ELOOP)

STEP	INSPECTION	RESULTS	ACTION
1	**PURPOSE: RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the snapshot data on the repair order.	—	Go to the troubleshooting procedure to perform the procedure from Step 1.

Troubleshooting Diagnostic Procedure (Without i-ELOOP)
Intention of troubleshooting procedure

- Step 1—6
  - — Perform an inspection of each signal transmission system.
- Step 7
  - — Perform the inspection of the part supplied power to the generator.
- Step 8
  - — Perform a unit inspection of the generator.
- Step 9—10
  - — Verify that the primary malfunction is resolved and there are no other malfunctions.

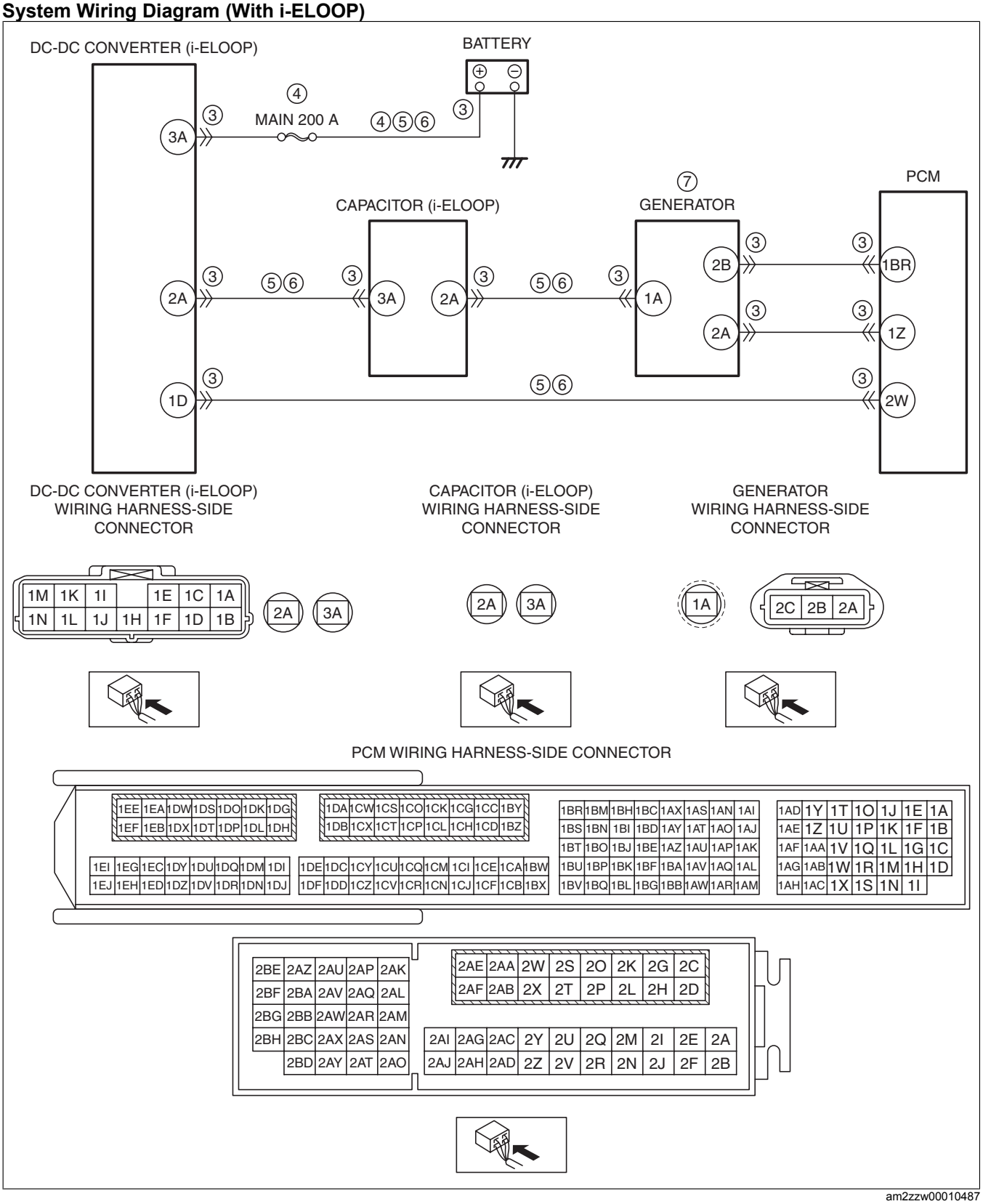
STEP	INSPECTION	RESULTS	ACTION
1	**PURPOSE: VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
2	**PURPOSE: VERIFY IF POOR CONNECTION OF EACH PART AFFECTS DIAGNOSTIC RESULTS** • Switch the ignition off. • Inspect the connection condition (part installation condition, connector connection condition) for the following parts: — Battery — Generator — PCM • Is the connection condition (part installation condition, connector connection condition) for each part normal?	Yes	Go to the next step.
		No	Connect each part or the connector correctly, then go to Step 9.
3	**PURPOSE: VERIFY IF CONNECTOR DAMAGE OF EACH PART AFFECTS DIAGNOSTIC RESULTS** • Disconnect the connector of the following parts. — Battery — Generator — PCM • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 9.
		No	Go to the next step.
4	**PURPOSE: INSPECT FUSE** • Remove the STR 200 A fuse. • Inspect the STR 200 A fuse. • Is there any malfunction?	Yes	If the fuse is blown: • Refer to the wiring diagram and verify whether or not there is a common connector between battery positive terminal and generator terminal 1A. **If there is a common connector:** — Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. — Repair or replace the malfunctioning part. **If there is no common connector:** — Repair or replace the wiring harness which has a short to ground. — Replace the malfunctioning fuse. If the fuse is damaged: • Replace the fuse. Go to Step 9.
		No	Reinstall the STR 200 A fuse, then go to the next step.

STEP	INSPECTION	RESULTS	ACTION
5	**PURPOSE: VERIFY IF SHORT TO GROUND IN GENERATOR CHARGE/DISCHARGE CIRCUIT AFFECTS DIAGNOSTIC RESULTS** • Verify that the battery, generator and PCM connectors are disconnected. • Inspect for continuity between generator terminal 1A (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between battery positive terminal and generator terminal 1A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 9.
		No	Go to the next step.
6	**PURPOSE: VERIFY IF OPEN CIRCUIT IN GENERATOR CHARGE/DISCHARGE CIRCUIT AFFECTS DIAGNOSTIC RESULTS** • Verify that the battery, generator and PCM connectors are disconnected. • Inspect for continuity between battery positive terminal (wiring harness-side) and generator terminal 1A (wiring harness-side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between battery positive terminal and generator terminal 1A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 9.
7	**PURPOSE: VERIFY IF DIAGNOSTIC RESULTS ARE AFFECTED BY INSUFFICIENT EXCITATION CURRENT TO GENERATOR DUE TO POWER SUPPLY PART MALFUNCTION** • Inspect the battery. (See BATTERY INSPECTION.) • Is there any malfunction?	Yes	Replace the battery, then go to Step 9. (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Go to the next step.
8	**PURPOSE: DETERMINE INTEGRITY OF GENERATOR** • Inspect the generator. (See GENERATOR INSPECTION [WITHOUT i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) • Is there any malfunction?	Yes	Replace the generator, then go to the next step. (See GENERATOR REMOVAL/INSTALLATION [WITHOUT i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].)
		No	Go to the next step.
9	**PURPOSE: VERIFICATION OF VEHICLE REPAIR COMPLETION** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Implement the repeatability verification procedure. (See Repeatability Verification Procedure (Without i-ELOOP).) • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) Go to the next step.
		No	Go to the next step.
10	**PURPOSE: VERIFY IF THERE IS ANY OTHER MALFUNCTION** • Is any other DTC or pending code stored?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	DTC troubleshooting completed.

Details On DTCs (With i-ELOOP)

DESCRIPTION	Generator system: Malfunction in voltage generated by generator	
DETECTION CONDITION	Determination conditions	• Any one of the following conditions is met: — Battery voltage is **11 V or less** and condition in which current sensor detects battery charging for specified period continues — Pre-charge mode is implemented and pre-charge mode is not completed even though specified period has elapsed.
	Preconditions	• All of the following conditions are met: — While engine is running — CAN communication condition related to current sensor is normal — Pre-charge mode is not operating — Bypass mode is not operating
	Malfunction determination period	• Pre-charge mode condition: **3 min** • Other condition: **5 s**
	Drive cycle	• 1
	Self test type	• CMDTC self test
	Sensor used	• Current sensor • BCM • PCM • Generator • DC-DC converter (i-ELOOP)
FAIL-SAFE FUNCTION	• Inhibits engine-stop by operating the i-stop function. (With i-stop) • Generator control is inhibited.	
VEHICLE STATUS WHEN DTCs ARE OUTPUT	• Flashes i-stop warning light (amber). (With i-stop) • A warning message is displayed on the center display. (With center display) • Illuminates charging system warning light. • If the vehicle continues to be driven while the DTC is detected the battery will be depleted. — A malfunction occurs with an electrical device and the vehicle stops. • The following vehicle conditions differ depending on the type of malfunction: — Vehicle shock may occur due to generator load. — Idling feel due to generator-stop may occur.	
POSSIBLE CAUSE	• Poor connection of the following parts: — Battery — Generator — PCM — Capacitor (i-ELOOP) — DC-DC converter (i-ELOOP) • Connector or terminal malfunction of the following parts: — Battery — Generator — PCM — Capacitor (i-ELOOP) — DC-DC converter (i-ELOOP) • MAIN 200 A fuse malfunction • Short to ground in wiring harness between the following terminals: — DC-DC converter (i-ELOOP) terminal 3A—Battery positive terminal — DC-DC converter (i-ELOOP) terminal 2A—Capacitor (i-ELOOP) terminal 3A — DC-DC converter (i-ELOOP) terminal 1D—PCM terminal 2W — Capacitor (i-ELOOP) terminal 2A—Generator terminal 1A • Open circuit in wiring harness between the following terminals: — DC-DC converter (i-ELOOP) terminal 3A—Battery positive terminal — DC-DC converter (i-ELOOP) terminal 2A—Capacitor (i-ELOOP) terminal 3A — DC-DC converter (i-ELOOP) terminal 1D—PCM terminal 2W — Capacitor (i-ELOOP) terminal 2A—Generator terminal 1A • Capacitor (i-ELOOP) malfunction • Generator malfunction • Current sensor malfunction • BCM malfunction • DC-DC converter (i-ELOOP) malfunction • PCM malfunction	

System Wiring Diagram (With i-ELOOP)



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Function Explanation (DTC Detection Outline) (With i-ELOOP)

- When the charge/discharge circuit for the power supplying the vehicle is normal, the vehicle will operate normally. In this diagnostic, a low vehicle supply voltage is detected even though the generator is generating power, a malfunction in the charge/discharge circuit is detected, and verification of vehicle malfunctions/safety assurance is performed.

Repeatability Verification Procedure (With i-ELOOP)

1. Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
2. Start the engine.
3. Leave for **180 s** while idling with no electrical load.
4. Leave for **30 s** while idling with high electrical load.

PID Item/Simulation Item Used In Diagnosis (With i-ELOOP)

- Not applicable

Function Inspection Using M-MDS (With i-ELOOP)

STEP	INSPECTION	RESULTS	ACTION
1	**PURPOSE: RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the snapshot data on the repair order.	—	Go to the next step.
2	**PURPOSE: VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**PURPOSE: VERIFY THE PRE-CHARGE MODE CONDITION AND DETERMINE THE MALFUNCTION LOCATION** • Start the engine. • Is the pre-charge warning displayed in the display?	Yes	Go to the next step.
		No	Go to the troubleshooting procedure to perform the procedure from Step 1.
4	**PURPOSE: VERIFY IF GENERATOR VOLTAGE IS FALSELY RECOGNIZED BY DTC RELATED CURRENT SENSOR** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is the PENDING CODE/DTC P058A:00 also present?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC P058A:00 [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) Go to the next step.
		No	Go to the next step.
5	**PURPOSE: VERIFY DTC** • Retrieve the PCM DTCs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) Go to the next step.
		No	Go to the next step.
6	**PURPOSE: VERIFY IF DIAGNOSTIC RESULT IS AFFECTED BY DTC RELATED TO DC-DC CONVERTER (i-ELOOP)** • Perform the DC-DC converter (i-ELOOP) DTC inspection using the M-MDS. (See DTC INSPECTION [i-ELOOP].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [i-ELOOP].) Go to the troubleshooting procedure to perform the procedure from Step 1.
		No	Go to the troubleshooting procedure to perform the procedure from Step 1.

Troubleshooting Diagnostic Procedure (With i-ELOOP)
Intention of troubleshooting procedure

- Step 1
  — Perform inspection of generator voltage.
- Step 2—6
  — Perform inspection of each signal transmission system.
- Step 7
  — Perform a unit inspection of the generator.
- Step 8—10

— Verify that primary malfunction is resolved and there are no other malfunctions.

STEP	INSPECTION	RESULTS	ACTION
1	**PURPOSE: DETERMINE THE MALFUNCTION CAUSE BY VERIFYING THE OUTPUT VOLTAGE OF GENERATOR TERMINAL 1A.** • Start the engine. • Measure the voltage at the generator terminal 1A (wiring harness-side). • Is the voltage **approx. 0 V**?	Yes	Perform the troubleshooting procedure for DTC P2503:00. (See DTC P2503:00 [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) Go to the next step.
		No	Go to the next step.
2	**PURPOSE: VERIFY IF POOR CONNECTION OF EACH PART AFFECTS DIAGNOSTIC RESULTS** • Switch the ignition off. • Remove the service plug. (See SERVICE PLUG DISCONNECTION/ CONNECTION [i-ELOOP].) • Inspect the connection condition (part installation condition, connector connection condition) for the following parts: — Battery — Generator — PCM — Capacitor (i-ELOOP) — DC-DC converter (i-ELOOP) • Is the connection condition (part installation condition, connector connection condition) for each part normal?	Yes	Go to the next step.
		No	Connect each part or the connector correctly, then go to Step 9.
3	**PURPOSE: VERIFY IF CONNECTOR DAMAGE OF EACH PART AFFECTS DIAGNOSTIC RESULTS** • Disconnect the connector of the following parts. — Battery — Generator — PCM — Capacitor (i-ELOOP) — DC-DC converter (i-ELOOP) • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 9.
		No	Go to the next step.
4	**PURPOSE: INSPECT FUSE** • Remove the MAIN 200 A fuse. • Inspect the MAIN 200 A fuse. • Is there any malfunction?	Yes	If the fuse is blown: • Refer to the wiring diagram and verify whether or not there is a common connector between DC-DC converter (i-ELOOP) terminal 3A and battery positive terminal. **If there is a common connector:** — Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. — Repair or replace the malfunctioning part. **If there is no common connector:** — Repair or replace the wiring harness which has a short to ground. — Replace the fuse. If the fuse is damaged: • Replace the fuse. Go to Step 9.
		No	Reinstall the MAIN 200 A fuse, then go to the next step.

STEP	INSPECTION	RESULTS	ACTION
5	**PURPOSE: VERIFY IF SHORT TO GROUND IN EACH WIRING HARNESS AFFECTS DIAGNOSTIC RESULTS** • Verify that the battery, generator, PCM, capacitor (i-ELOOP), and DC-DC converter (i-ELOOP) connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side) and body ground: — DC-DC converter (i-ELOOP) terminal 3A — DC-DC converter (i-ELOOP) terminal 2A — DC-DC converter (i-ELOOP) terminal 1D — Generator terminal 1A • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • DC-DC converter (i-ELOOP) terminal 3A—Battery positive terminal • DC-DC converter (i-ELOOP) terminal 2A—Capacitor (i-ELOOP) terminal 3A • DC-DC converter (i-ELOOP) terminal 1D—PCM terminal 2W • Capacitor (i-ELOOP) terminal 2A—Generator terminal 1A **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 9.
		No	Go to the next step.
6	**PURPOSE: VERIFY IF OPEN CIRCUIT IN EACH WIRING HARNESS AFFECTS DIAGNOSTIC RESULTS** • Verify that the battery, generator, PCM, capacitor (i-ELOOP), and DC-DC converter (i-ELOOP) connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side): — DC-DC converter (i-ELOOP) terminal 3A—Battery positive terminal — DC-DC converter (i-ELOOP) terminal 2A—Capacitor (i-ELOOP) terminal 3A — DC-DC converter (i-ELOOP) terminal 1D—PCM terminal 2W — Capacitor (i-ELOOP) terminal 2A—Generator terminal 1A • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • DC-DC converter (i-ELOOP) terminal 3A—Battery positive terminal • DC-DC converter (i-ELOOP) terminal 2A—Capacitor (i-ELOOP) terminal 3A • DC-DC converter (i-ELOOP) terminal 1D—PCM terminal 2W • Capacitor (i-ELOOP) terminal 2A—Generator terminal 1A **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 9.
7	**PURPOSE: DETERMINE INTEGRITY OF GENERATOR** • Inspect the generator. (See GENERATOR INSPECTION [WITH i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) • Is there any malfunction?	Yes	Replace the generator, then go to the next step. (See GENERATOR REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].)
		No	Go to the next step.
8	**PURPOSE: VERIFY CONDITIONS OF BATTERY** • Inspect the battery. (See BATTERY INSPECTION.)	—	Follow the inspection instructions, then go to the next step.

STEP	INSPECTION	RESULTS	ACTION
9	**PURPOSE: VERIFICATION OF VEHICLE REPAIR COMPLETION** • Always reconnect all disconnected connectors. • Reinstall the service plug. (See SERVICE PLUG DISCONNECTION/ CONNECTION [i-ELOOP].) • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Implement the repeatability verification procedure. (See Repeatability Verification Procedure (With i-ELOOP).) • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) Go to the next step.
		No	Go to the next step.
10	**PURPOSE: VERIFY IF THERE IS ANY OTHER MALFUNCTION** • Is any other DTC or pending code stored?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	DTC troubleshooting completed.